

基本概念练习题

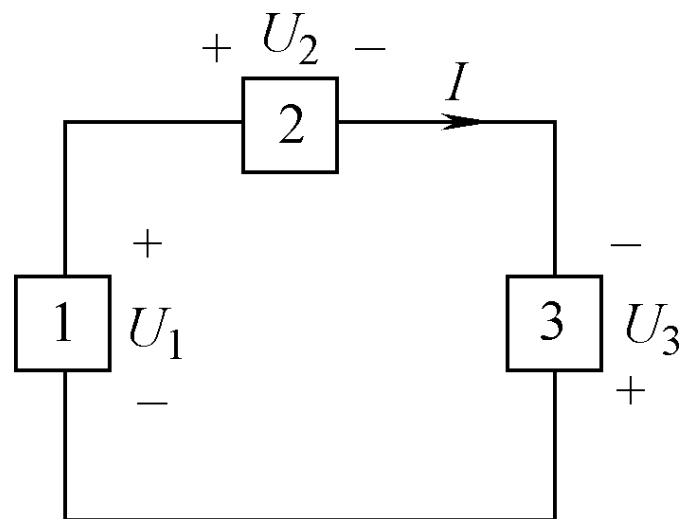
【例1】 如图所示的电路中，已知 $I = 2\text{A}$ ， $U_1 = 10\text{V}$ ， $U_2 = 6\text{V}$ ， $U_3 = -4\text{V}$ ，试问哪些元件是电源？哪些元件是负载？

第一种方法：用实际方向判断电源与负载。

$U_1 \rightarrow$ 电源，发出功率

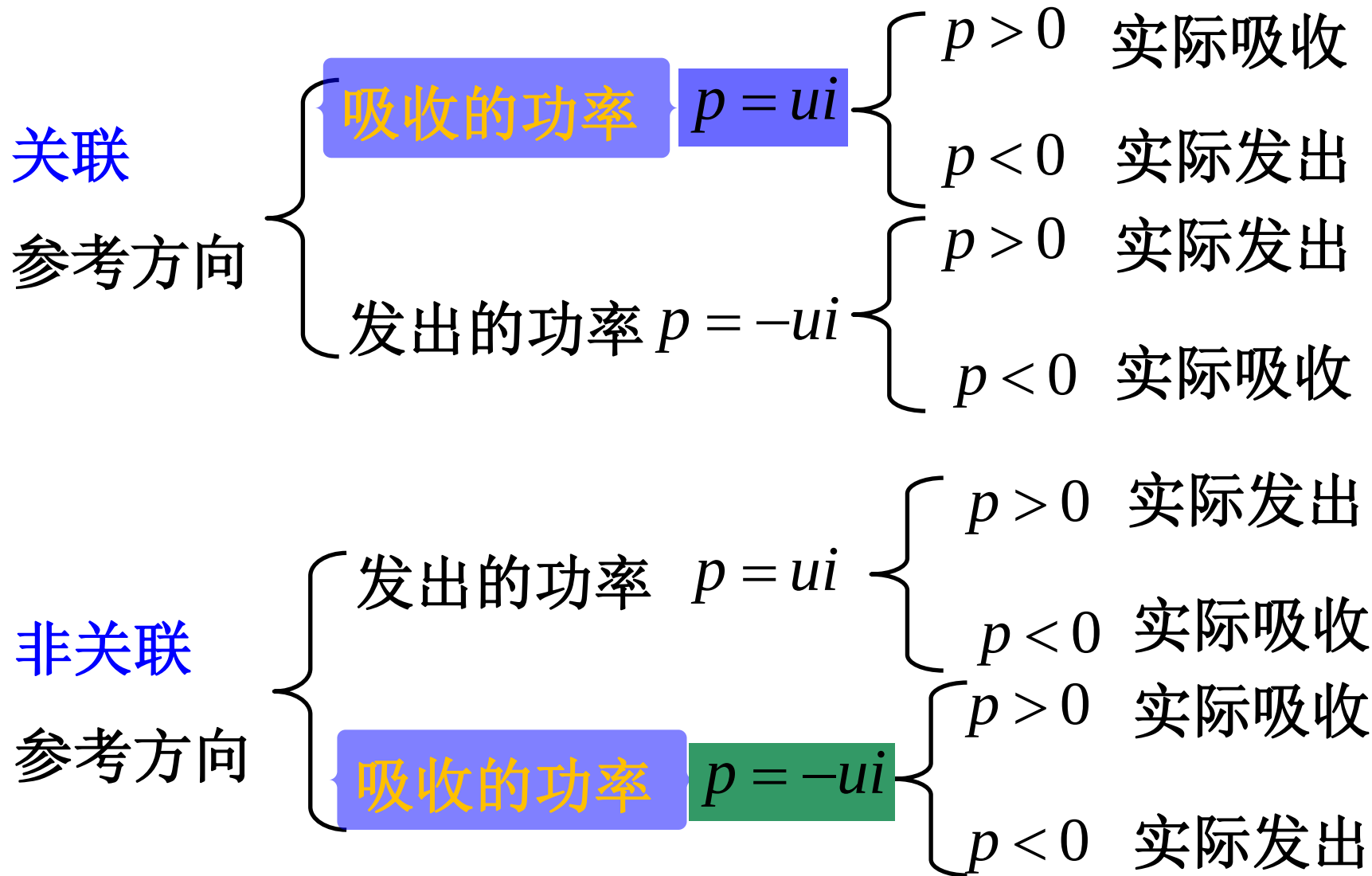
$U_2 \rightarrow$ 负载，吸收功率

$U_3 \rightarrow$ 负载，吸收功率



第二种方法：计算功率判断电源与负载。

功率与参考方向之间的关系



【例1】如图所示的电路中，已知 $I = 2\text{A}$ ， $U_1 = 10\text{V}$ ，

$U_2 = 6\text{V}$ ， $U_3 = -4\text{V}$ ，试问哪些元件是电源？哪些

元件是负载？

第二种方法：计算功率 P 。

U_1 发出功率为

$$P = U_1 I = 10 \times 2 = 20\text{W} \quad (\text{实际发出})$$

U_1 吸收功率为

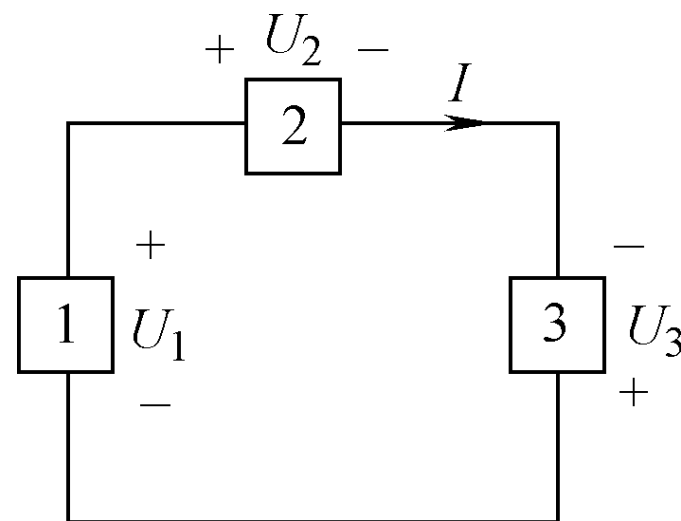
$$P = -U_1 I = -10 \times 2 = -20\text{W} \quad (\text{实际发出})$$

U_2 发出功率为 $P = -U_2 I = -6 \times 2 = -12\text{W}$ （实际吸收）

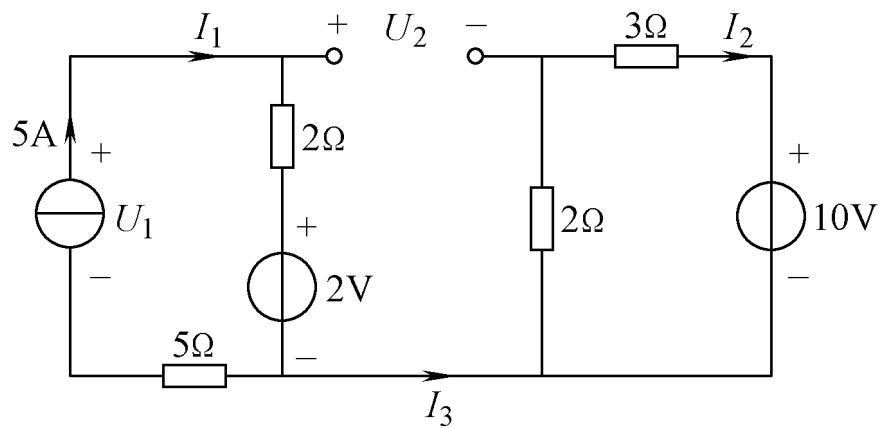
U_2 吸收功率为 $P = U_2 I = 6 \times 2 = 12\text{W}$ （实际吸收）

U_3 发出功率为 $P = U_3 I = -4 \times 2 = -8\text{W}$ （实际吸收）

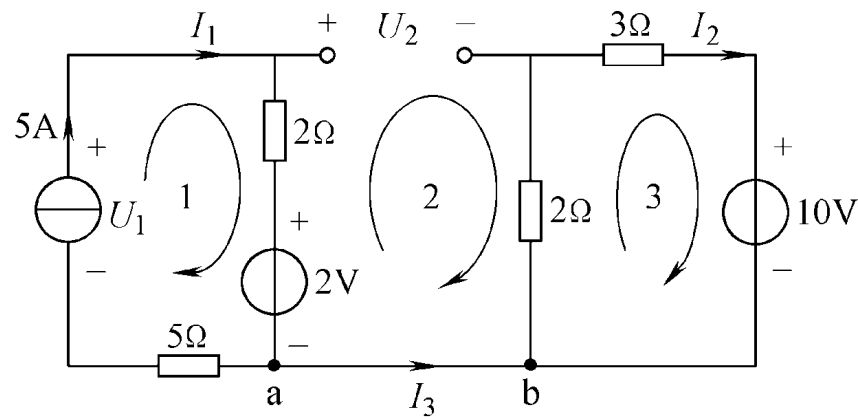
U_3 吸收功率为 $P = -U_3 I = 4 \times 2 = 8\text{W}$ （实际吸收）



【例2】 电路如图a所示，求电流 I_1 、 I_2 、 I_3 和电压 U_1 、 U_2 。



a)



b)

【解】 设三个回路的绕行参考方向如图b所示。

$$I_1 = 5\text{A} \quad I_2 = -\frac{10}{2+3}\text{A} = -2\text{A} \quad I_3 = 0$$

$$U_1 = 2I_1 + 2 + 5I_1 = 7I_1 + 2 = (7 \times 5 + 2)\text{V} = 37\text{V}$$

$$U_2 = 2I_1 + 2 + 2I_2 = [2 \times 5 + 2 + 2 \times (-2)]\text{V} = 8\text{V}$$

【例3】求图示电路中开关S打开和闭合时a点的电位值。

【解】开关S断开，三个电阻串联。电路两端点电压为

$$U = (12 - (-12))\text{V} = 24\text{V}$$

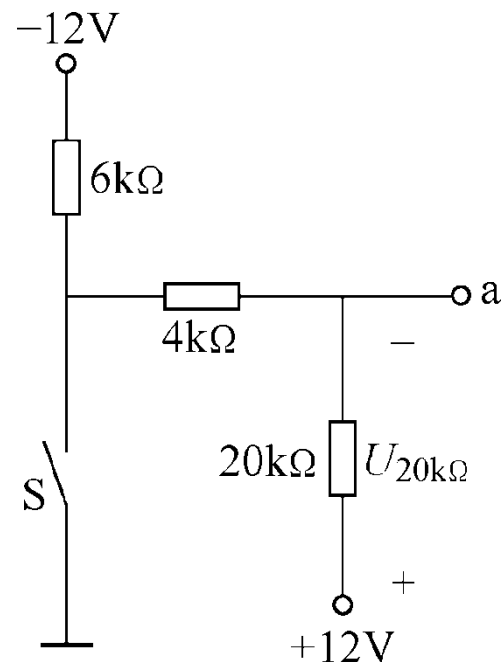
20kΩ 电阻两端的电压为

$$U_{20\text{k}\Omega} = 20 \times \frac{24}{6 + 4 + 20} \text{V} = 16\text{V}$$

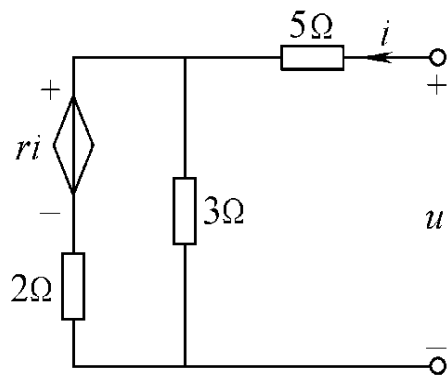
$$V_a = 12 - U_{20\text{k}\Omega} = (12 - 16)\text{V} = -4\text{V}$$

开关S闭合后，有

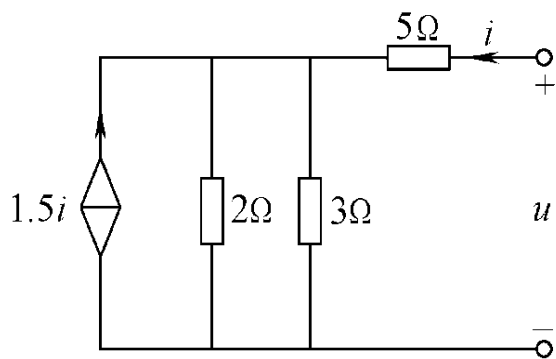
$$V_a = \frac{12}{4 + 20} \times 4\text{V} = 2\text{V}$$



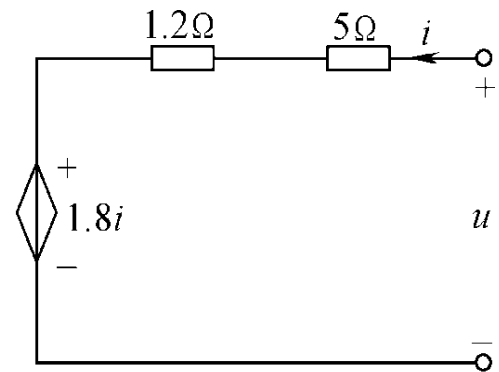
【例4】 在图a所示的电路中，已知转移电阻系数 $r = 3\Omega$ 求一端口网络的等效电阻。



a)



b)



c)

【解】 首先将图a中的串联支路等效变换为图b中的并联支路，依次化简如图c所示。

$$u = (5 + 1.2 + 1.8)i = 8i \quad \Rightarrow \quad R = \frac{u}{i} = 8\Omega$$

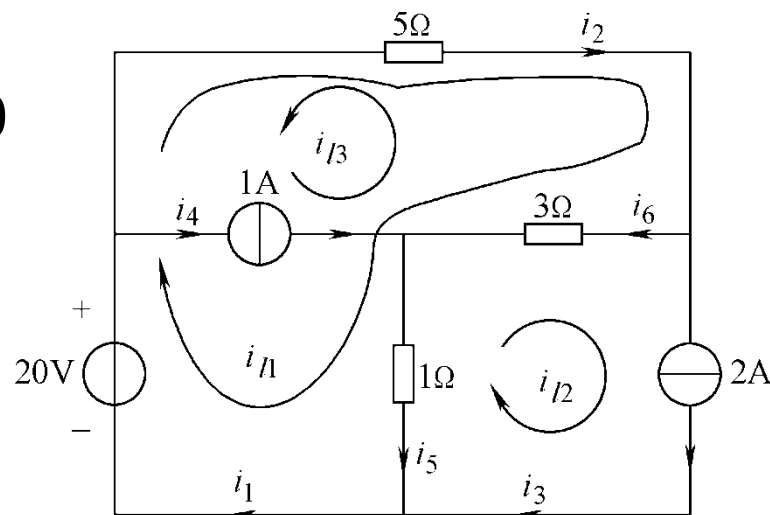
【例5】 用回路电流法求各支路电流。

【解】 选定三个回路电流 i_{l1} 、 i_{l2} 和 i_{l3} 的参考方向
如图所示。列写回路电流方程

$$(5 + 3 + 1)i_{l1} - (1 + 3)i_{l2} - (5 + 3)i_{l3} = 20$$

$$i_{l2} = 2A$$

$$i_{l3} = 1A \quad i_{l1} = 4A$$



各支路电流分别为

$$i_1 = i_{l1} = 4A$$

$$i_2 = i_{l1} - i_{l3} = 3A$$

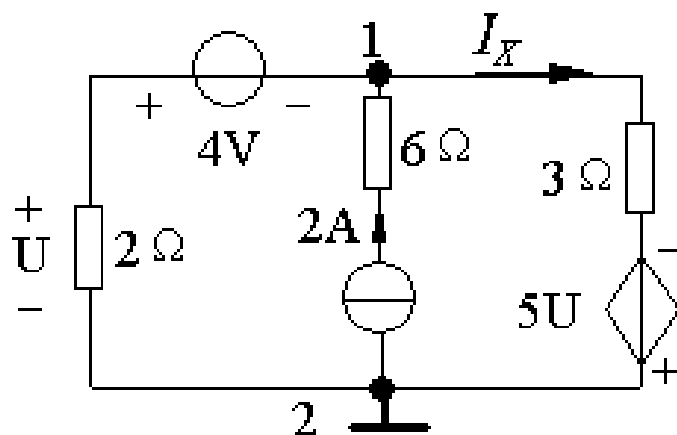
$$i_3 = i_{l2} = 2A$$

$$i_4 = i_{l3} = 1A$$

$$i_5 = i_{l1} - i_{l2} = 2A$$

$$i_6 = i_{l1} - i_{l2} - i_{l3} = 1A$$

【例6】 已知电路如图所示，用节点电压法求 I_X 。



解：

$$\begin{cases} U_1 = \frac{2 - 4/2 - 5U/3}{1/2 + 1/3} \\ U = 4 + U_1 \end{cases} \quad \begin{cases} U_1 = -\frac{8}{3} \text{ V} \\ U = \frac{4}{3} \text{ V} \end{cases}$$

$$I_X = \frac{U_1 + 5U}{3} = \frac{4}{3} \text{ A}$$

【例7】 试用结点法求电压 U_X

【解】 $u_{n2} = 1$

$$\left(\frac{1}{1} + \frac{1}{2}\right)u_{n1} - \frac{1}{1}u_{n2} = 1$$

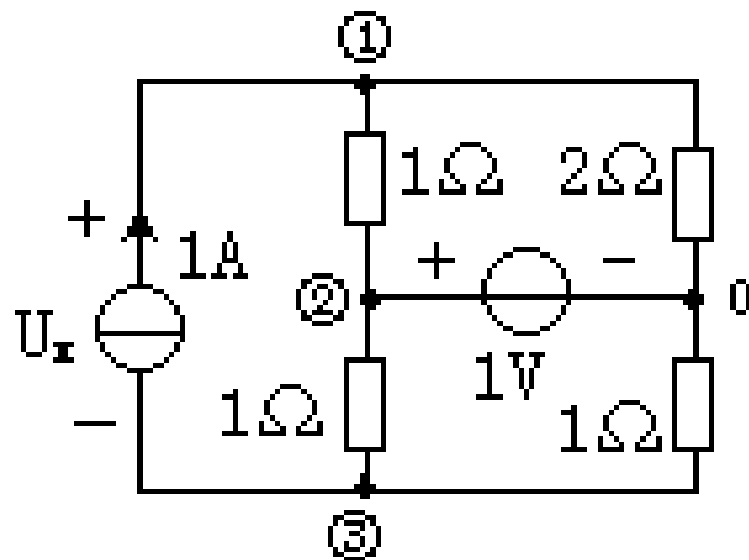
$$-\frac{1}{1}u_{n2} + \left(\frac{1}{1} + \frac{1}{1}\right)u_{n3} = -1$$

$$u_{n1} = 1.33V$$

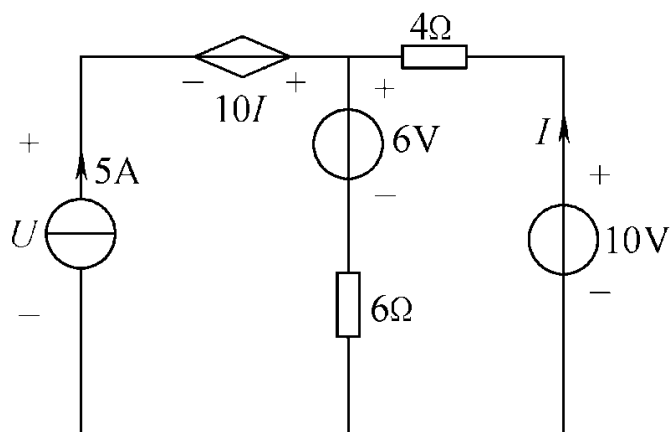
$$-1 + 2u_{n3} = -1$$

$$u_{n3} = 0$$

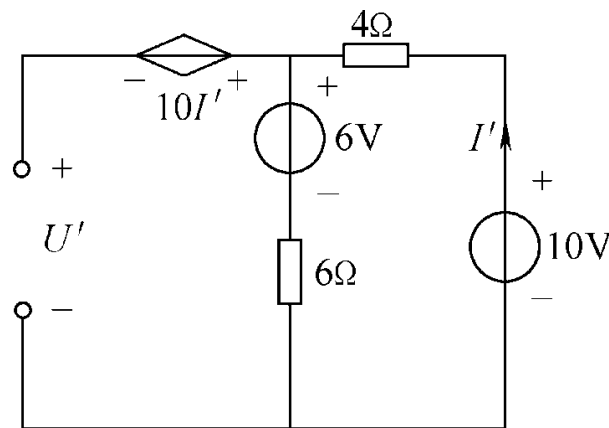
$$u_X = u_{n1} - u_{n3} = 1.33V$$



【例8】 电路如图a所示，试用叠加定理求 U 和 I 。



a)



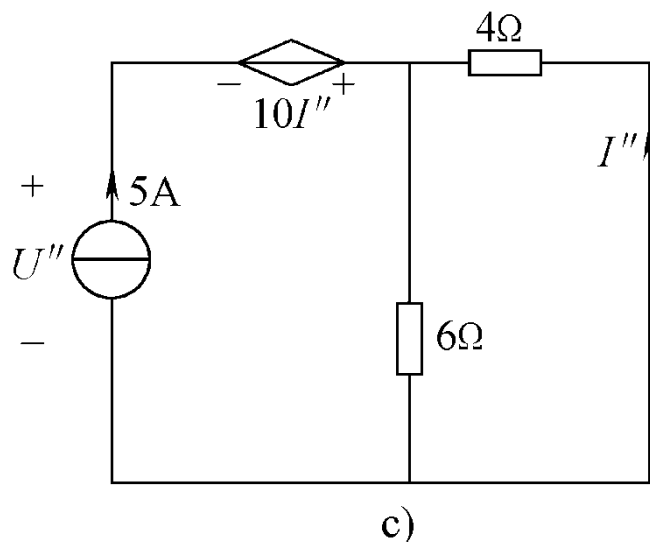
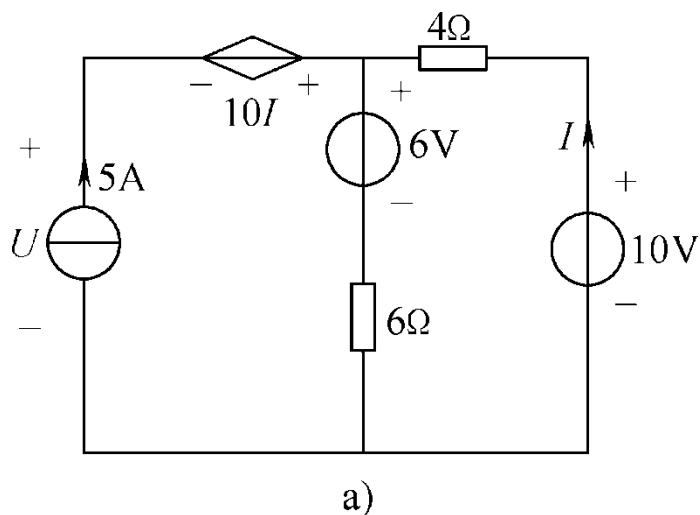
b)

【解】 (1) 将电源分成组，即当6V电压源和10V电压源共同作用时，5A电流源用开路代替，电路如图b所示。根据KVL和欧姆定律，得

$$I' = \frac{10 - 6}{6 + 4} \text{ A} = 0.4 \text{ A}$$

$$U' = -10I' - 4I' + 10 = 4.4 \text{ V}$$

(2) 5A电流源单独作用时的电路如图c所示，根据分流公式得



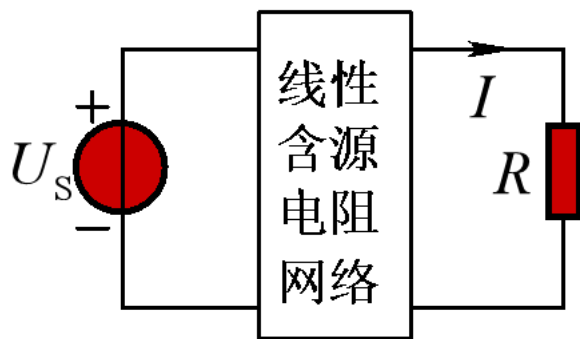
$$I'' = -\frac{6}{4+6} \times 5A = -3A$$

$$U'' = -10I'' - 4I'' = 42V$$

(3) 利用叠加定理得 $I = I' + I'' = (0.4 - 3)A = -2.6A$

$$U = U' + U'' = 46.4V$$

【例9】 已知电路如图所示，当 $U_s = 10V$ 时 $I = 6A$



$U_s = 15V$ 时 $I = 7A$

$I = 10A$ 时 $U_s = ?$

【解】 根据齐性定理和叠加定理

$$I = I' + I'' = I' + kU_s$$

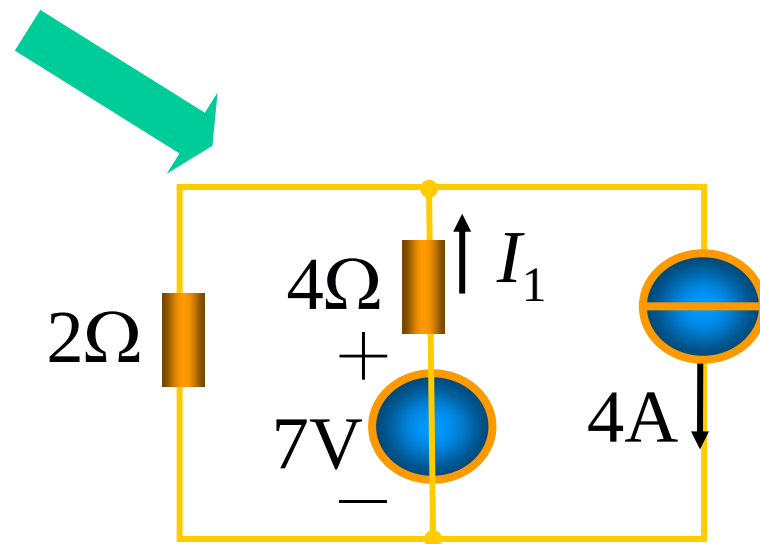
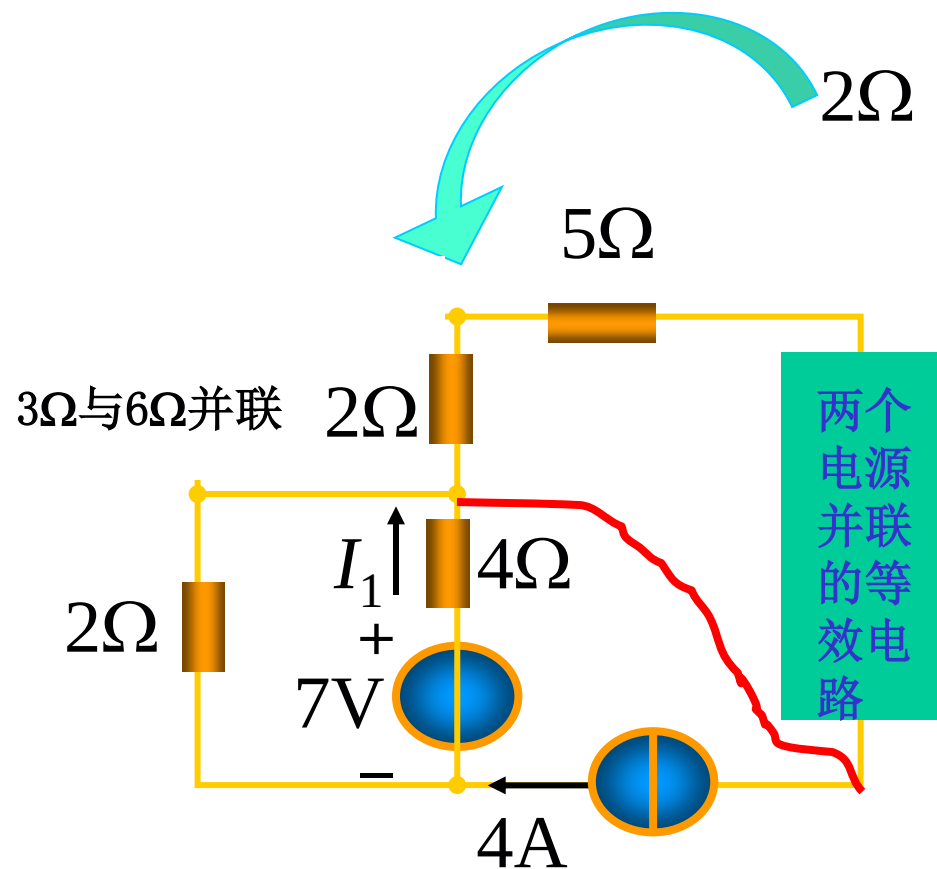
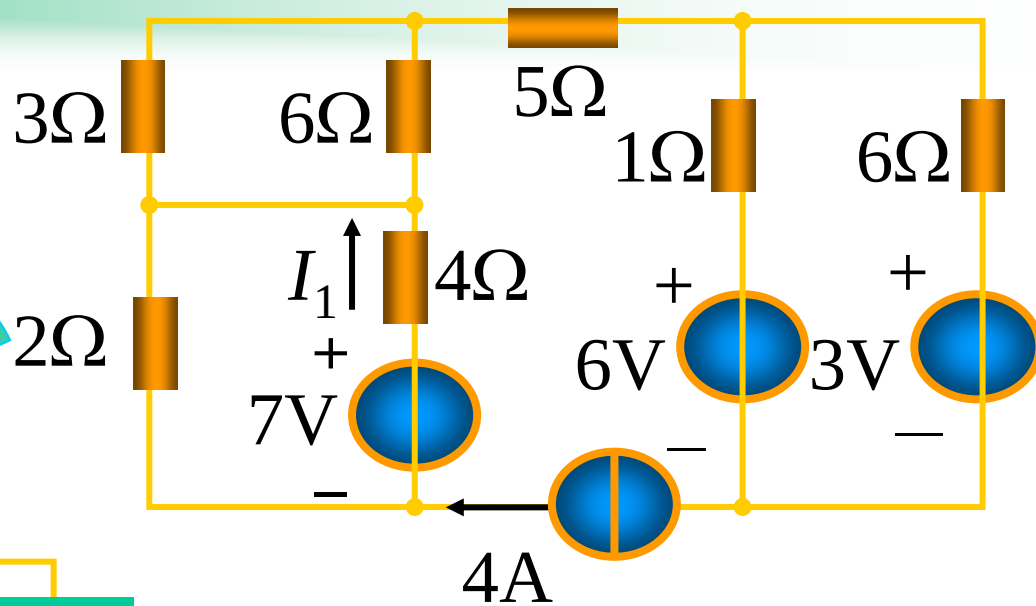
$$\begin{cases} 6A = I' + k \times 10V \\ 7A = I' + k \times 15V \end{cases} \Rightarrow \begin{cases} k = 0.2S \\ I' = 4A \end{cases}$$

$$I = 4A + 0.2S \times U_s$$

当 $I = 10A$ 得 $U_s = 30V$

【例10】 求电流 I_1

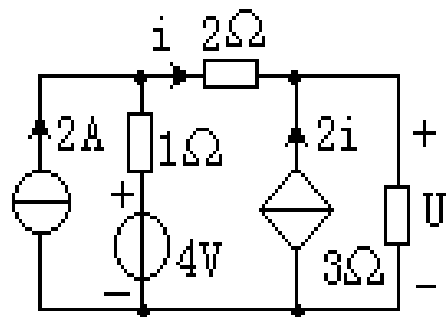
【解】 用替代定理:



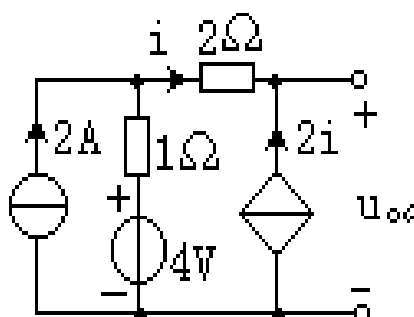
应用叠加原理

$$I_1 = \frac{7}{6} + \frac{2 \times 4}{2 + 4} = \frac{15}{6} = 2.5\text{A}$$

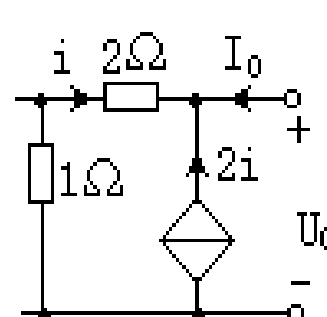
【例11】 电路如图(a)所示，试应用戴维宁定理求电压 U 。



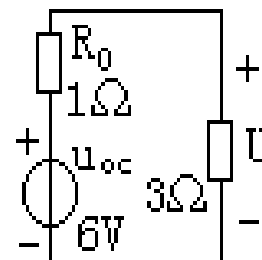
(a)



(b)



(c)



(d)

解 求开路电压 u_{oc} ：将图 (a)中 3Ω 电阻断开，则得图 (b)电路。

$$\because i = -2i \therefore i = 0 \text{ 则有 } u_{oc} = 4 + 2 \times 1 = 6V$$

用加压求流法求等效电阻 R_0 ：令网络内所有独立电源为零值，电路如图 (c)所示。

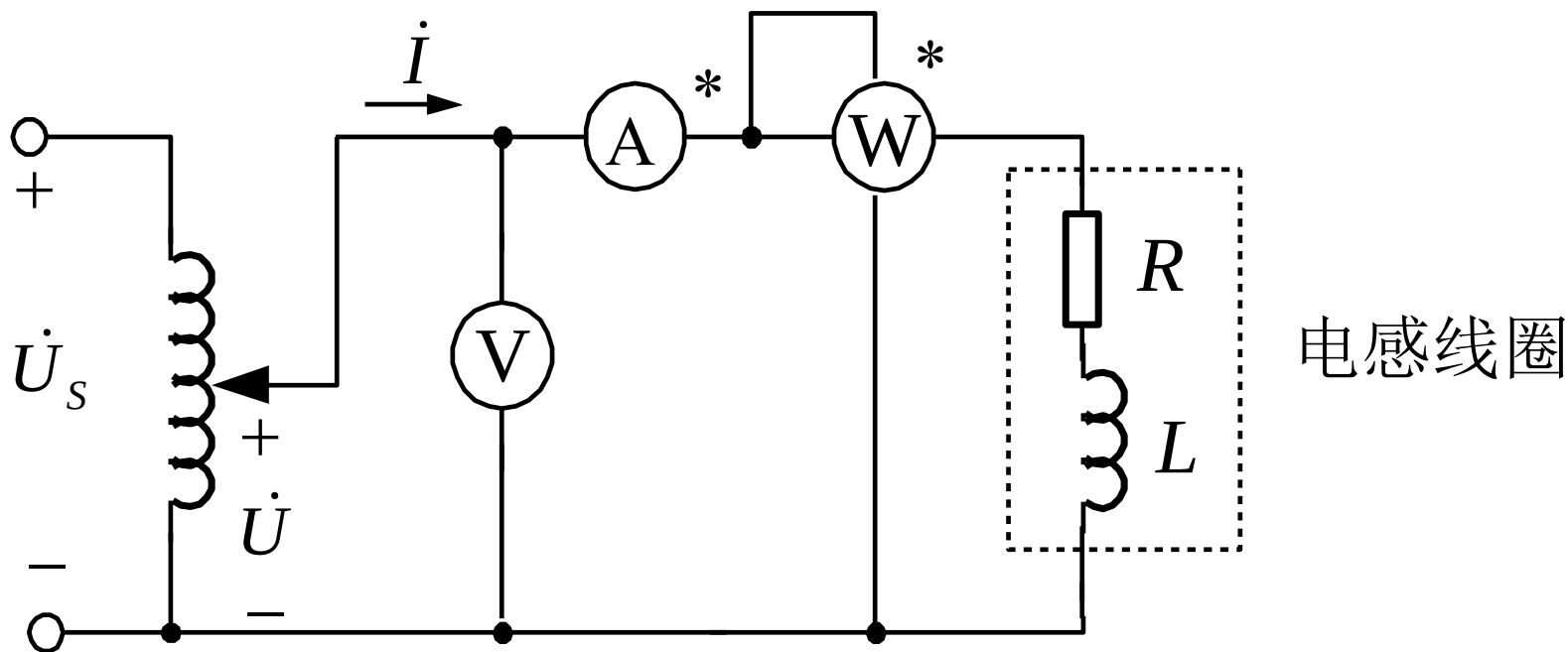
$$\because U_0 = -i \times (1 + 2) \therefore i = -\frac{1}{3}U_0 \quad \rightarrow \quad R_0 = \frac{U_0}{I_0} = 1\Omega$$

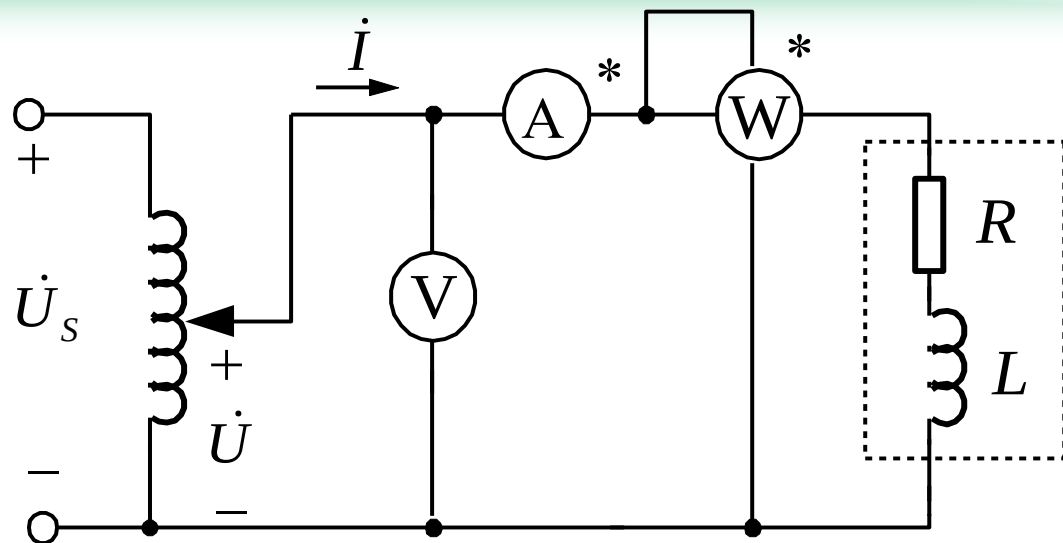
$$I_0 = -i - 2i = -3i = U_0$$

$$U = \frac{3}{R_0 + 3} \cdot u_{oc} = \frac{3}{1 + 3} \times 6 = 4.5V$$

【例12】

已知电路如图所示，电压表的读数为50V，电流表的读数为1A，功率表的读数为30W，电源的频率为50Hz。试求R、L的数值、无功功率、视在功率。





电压表 = 50V

电流表 = 1A

功率表 = 30W

解: 1.

$$Z = R + jX_L = |Z| \angle \varphi$$

$$\because P = UI \cos \varphi$$

$$|Z| = \frac{U}{I} = \frac{50}{1} = 50 \Omega$$

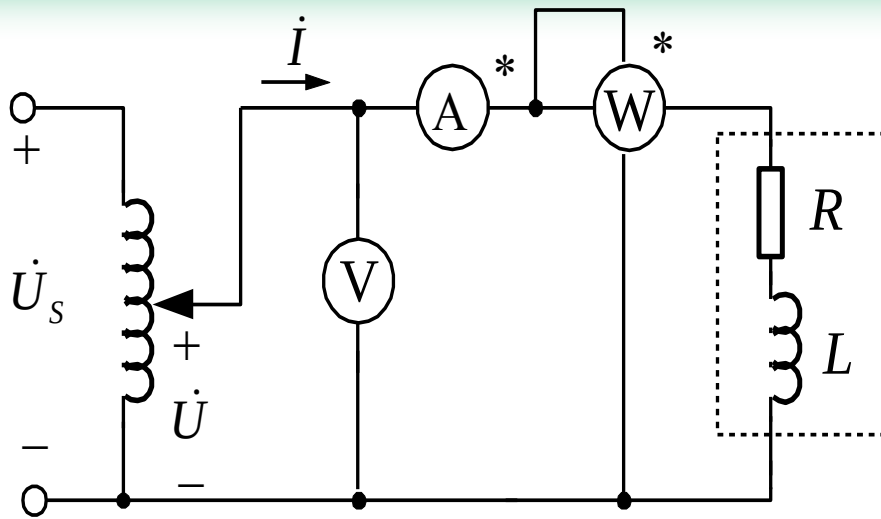
$$\therefore \cos \varphi = \frac{P}{UI} = \frac{30}{50 \times 1} = 0.6$$

$$Z = R + jX_L = |Z| \angle \varphi = 50 \angle 53.13^\circ \Omega$$

$$\varphi = 53.13^\circ$$

$$Z = 50 \angle 53.13^\circ = (30 + j40) \Omega$$

$$R = 30 \Omega, \quad L = \frac{X_L}{\omega} = \frac{40}{314} = 127 \text{ mH}$$



电压表 = 50V

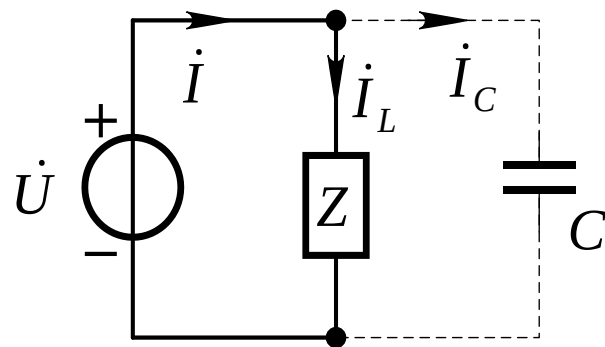
电流表 = 1A

功率表 = 30W

$$2. Q = UI \sin \varphi = 50 \times 1 \times \sin 53.13^\circ = 40 \text{ 乏}$$

$$3. S = UI = \sqrt{P^2 + Q^2} = 50 \times 1 = 50 \text{ VA}$$

【例13】在如图所示电路中，已知 $U = 220\text{V}$ ， $f = 50\text{Hz}$ ，感性负载的功率为 40W ，额定电流为 0.4A 。试求（1）电路的功率因数，电感 L 和电感上的电压；（2）若要将电路的功率因数提高到 0.95 ，需要并联多大电容？（3）并联电容后电源的总电流为多少？电源提供的无功功率为多少？



【解】（1）因为 $U = 220\text{V}$

$$\cos \varphi = \cos \varphi_L = \frac{P}{UI_N} = \frac{40}{220 \times 0.4} = 0.45$$

$$|Z| = \frac{U}{I_N} = \frac{220}{0.4} = 550\Omega \quad \varphi_L = \arccos \cos \varphi_L = 63^\circ$$

$$Z_L = |Z| \angle \varphi_L = 550 \angle 63^\circ = (250 + j490)\Omega$$

$$R = 250\Omega \quad X_L = 490\Omega$$

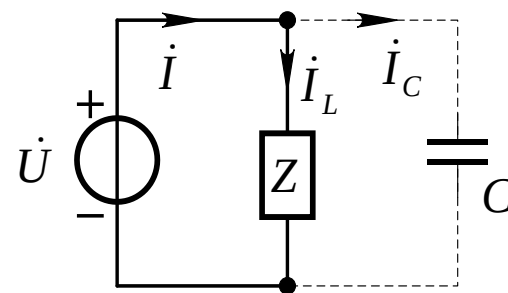
$$L = \frac{X_L}{\omega} = \frac{490}{314} \text{H} = 1.56\text{H}$$

$$U_L = X_L I_N = 490 \times 0.4\text{V} = 196\text{V}$$

(2) 并电容后, 电路的功率因数为0.95

$$\varphi = \arccos 0.95 = 18.2^\circ$$

由公式有 $C = \frac{P}{\omega U^2} (\tan \varphi_L - \tan \varphi)$



或者:

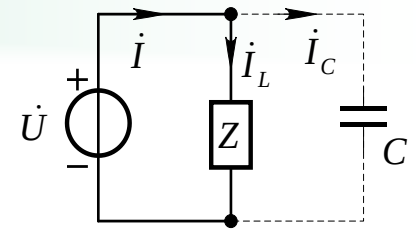
$$= \frac{40}{2\pi \times 50 \times 220^2} (\tan 63^\circ - \tan 18.2^\circ) \approx 4.3 \mu\text{F}$$

$$Q_L = I_L^2 X_L = 0.4^2 \times 490 = 78.4 \text{ var}$$

$$Q' = P \tan 18.2^\circ = 13.1 \text{ var}$$

$$Q_C = Q' - Q_L = 13.1 - 78.4 = -65.2 \text{ var}$$

$$C = \frac{Q_C}{\omega U^2} = \frac{65.2}{314 \times 220^2} = 4.3 \mu\text{F}$$

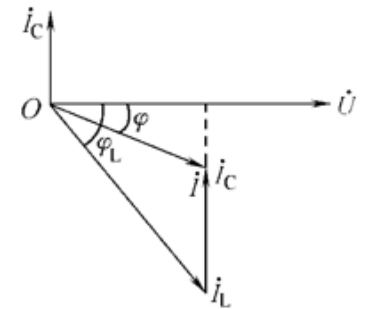


(3) 并联电容后，电源的总电流为

$$I = \frac{P}{U \cos \varphi} = \frac{40}{220 \times 0.95} \text{ A} = 0.191 \text{ A}$$

电源提供的无功功率为

$$Q = P \tan 18.2^\circ = 13.1 \text{ var}$$

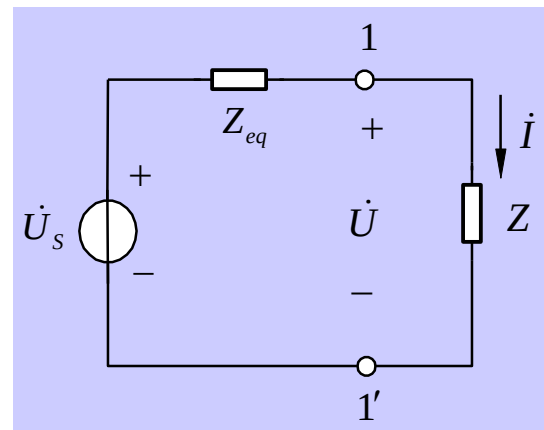
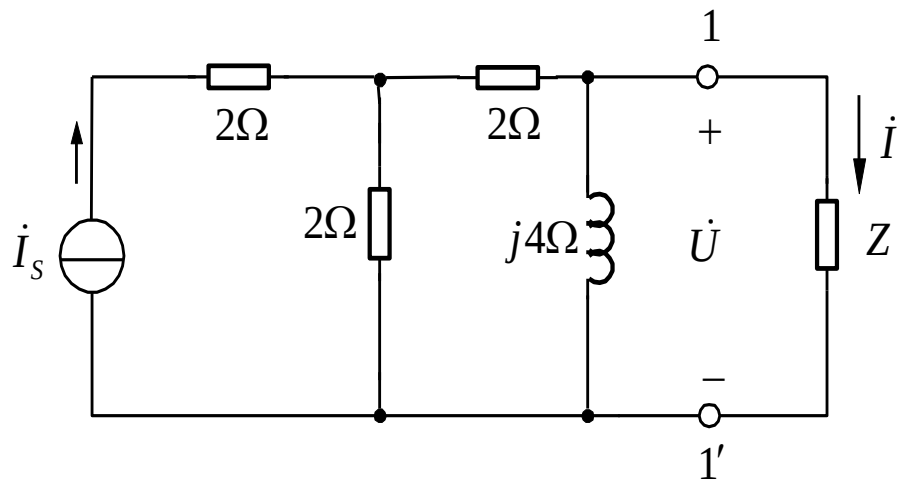


$$Q = Q_L + Q_C = 78.4 \text{ var} - 65.24 \text{ var} = 13.1 \text{ var}$$

$$Q = UI \sin \varphi = 220 \times 0.191 \times \sin 18.2 = 13.1 \text{ var}$$

【例14】电路如图所示，已知 $\dot{I}_s = 2\angle 0^\circ \text{ A}$ 。

求最佳匹配时获得的最大功率。



解：先求得一端口的戴维南等效电路

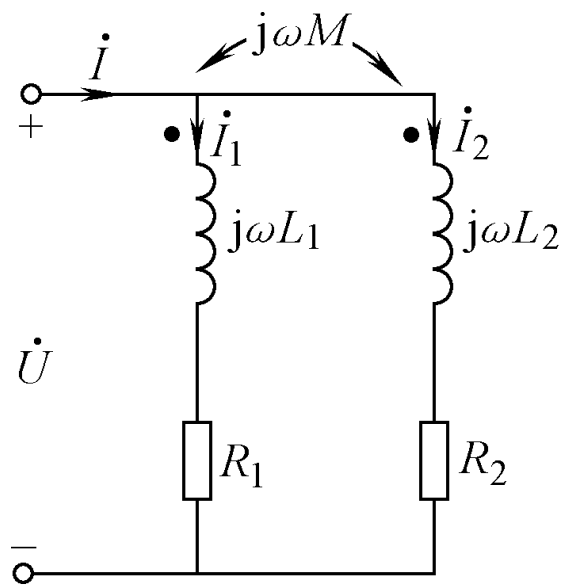
$$\begin{aligned}\dot{U}_{OC} &= \dot{U}_s = \frac{2 \times 2\angle 0^\circ}{2 + 2 + j4} \times j4 = \frac{j16}{5.65\angle 45^\circ} \\ &= 2.83\angle 45^\circ \text{ V}\end{aligned}$$

$$Z = Z_{eq}^* = 2 - j2 \Omega$$

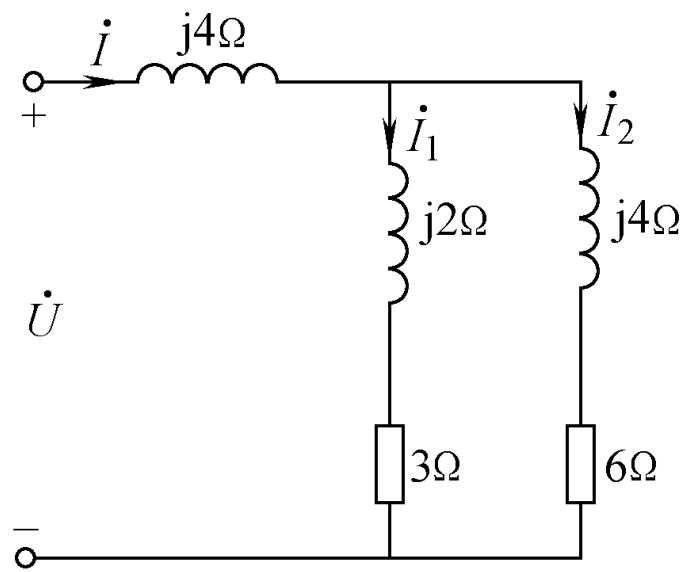
$$P_{\max} = \frac{U_{OC}^2}{4R_{eq}} = \frac{2.83^2}{4 \times 2} = 1 \text{ W}$$

$$Z_{eq} = \frac{4 \times j4}{4 + j4} = \frac{j16}{5.65\angle 45^\circ} = 2.83\angle 45^\circ = 2 + j2 \Omega$$

【例15】在图a)电路中，已知 $R_1 = 3\Omega$, $R_2 = 6\Omega$, $\omega L_1 = 6\Omega$, $\omega L_2 = 8\Omega$, $\omega M = 4\Omega$, $\dot{U} = 10\angle 0^\circ \text{ V}$ 。试用互感消去法求 $\dot{I}_1$, $\dot{I}_2$ 和输入阻抗 Z_i 。



a) 原电路



b) 等效电路

【解】将图a)电路中的互感消去，其等效电路如图 b)所示。

由图b)得

$$Z_1 = (3 + j2)\Omega = 3.6\angle 33.6^\circ \Omega$$

$$Z_2 = (6 + j4)\Omega = 7.2\angle 33.6^\circ \Omega$$

$$Z_1 // Z_2 = \frac{(3 + j2)(6 + j4)}{3 + j2 + 6 + j4} \Omega = \frac{25.92\angle 67.2^\circ}{9 + j6} \Omega$$

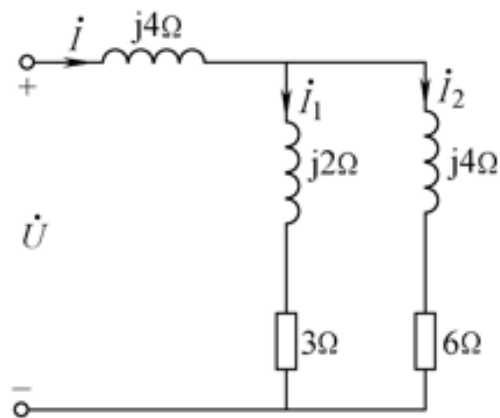
$$= 2.4\angle 33.6^\circ \Omega = (2 + j1.32) \Omega$$

输入阻抗为 $Z_i = j4 + Z_1 // Z_2 = (j4 + 2 + j1.32)\Omega = (2 + j5.32)\Omega$

$$\dot{I} = \frac{\dot{U}}{Z_i} = \frac{10\angle 0^\circ}{2 + j5.32} \text{ A} = 1.76\angle -69.4^\circ \text{ A}$$

$$\begin{aligned} \dot{I}_1 &= \frac{6 + j4}{3 + j2 + 6 + j4} \dot{I} = \frac{7.2\angle 33.6^\circ}{10.8\angle 33.6^\circ} \times 1.76\angle -69.4^\circ \text{ A} \\ &= 1.16\angle -69.4^\circ \text{ A} \end{aligned}$$

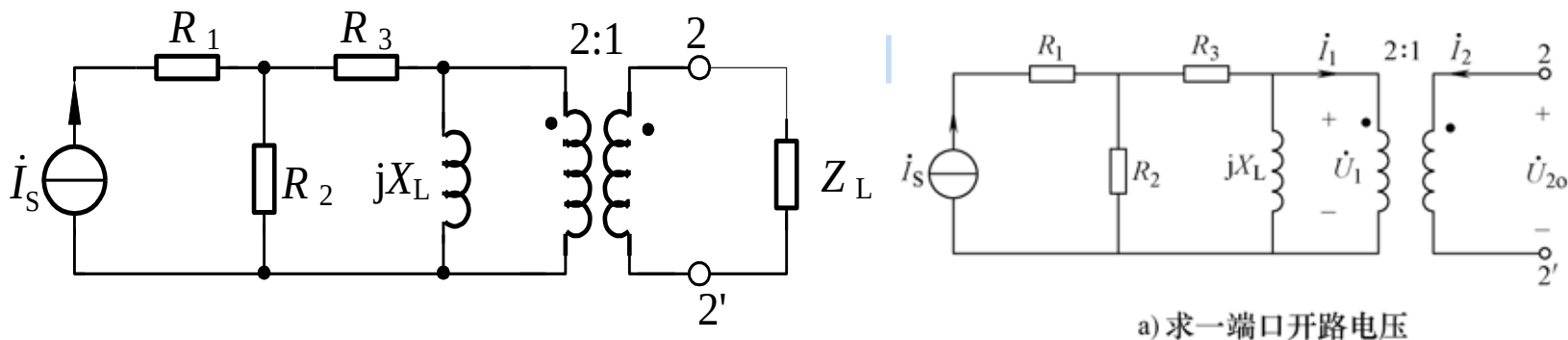
$$\begin{aligned} \dot{I}_2 &= \frac{3 + j2}{3 + j2 + 6 + j4} \dot{I} = \frac{3.6\angle 33.6^\circ}{10.8\angle 33.6^\circ} \times 1.76\angle -69.4^\circ \text{ A} \\ &= 0.58\angle -69.4^\circ \text{ A} \end{aligned}$$



b) 等效电路

【例16】在图7.31所示的电路中，已知 $\dot{I}_S = 4\angle 0^\circ \text{ A}$ ， $R_1 = R_2 = R_3 = 2\Omega$ ， $jX_L = j4\Omega$ 。试求：

- (1) 22'端口电路的戴维南等效电路；
- (2) 最佳匹配时负载 Z_L 上获得的最大功率。



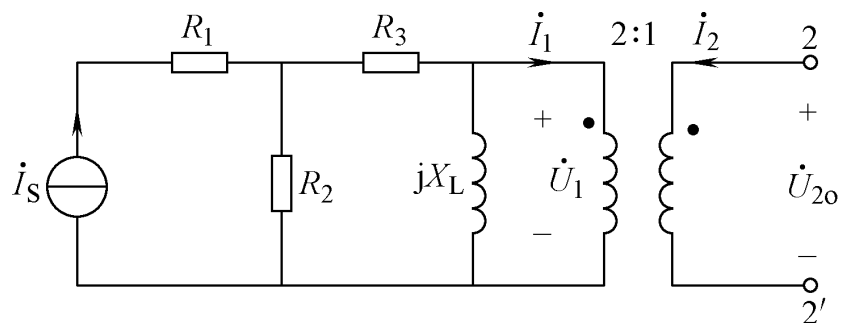
a) 求一端口开路电压

【解】 (1) 求开路电压的一端口电路如图a) 所示。

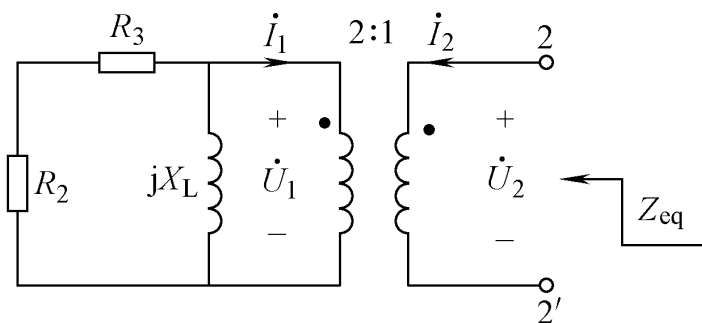
$$\dot{U}_{20} = \frac{1}{K} \dot{U}_1 = \frac{1}{2} \dot{U}_1 \quad \text{由于 } \dot{I}_2 = 0, \text{ 所以 } \dot{I}_1 = -\frac{1}{K} \dot{I}_2 = 0, \text{ 则}$$

$$\dot{U}_1 = jX_L \frac{R_2}{R_2 + R_3 + jX_L} \dot{I}_S = j4 \times \frac{2}{2 + 2 + j4} \times 4\angle 0^\circ = 4\sqrt{2}\angle 45^\circ \text{ V}$$

$$\dot{U}_{20} = \frac{1}{2} \dot{U}_1 = \frac{1}{2} \times 4\sqrt{2}\angle 45^\circ \text{ V} = 2\sqrt{2}\angle 45^\circ \text{ V}$$



a) 求一端口开路电压



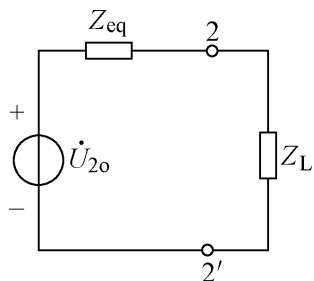
b) 求等效阻抗

由图b), 用外加电压法得

$$Z_{eq} = \frac{\dot{U}_2}{\dot{I}_2} = \frac{\frac{\dot{U}_1}{K}}{-K\dot{I}_1} = \frac{1}{K^2} \frac{\dot{U}_1}{(-\dot{I}_1)} = \frac{1}{K^2} \frac{[(R_2 + R_3) // jX_L](-\dot{I}_1)}{(-\dot{I}_1)} = \frac{1}{4} \times 4 // j4\Omega = \left(\frac{1}{2} + j\frac{1}{2}\right)\Omega$$

图a)的 22' 端口的戴维宁等效电路如下图所示。

当 $Z_L = Z_{eq}^* = \left(\frac{1}{2} - j\frac{1}{2}\right)\Omega$ 时,

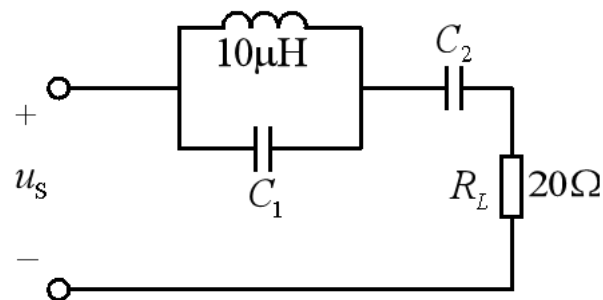


$$P_L = \frac{U_{2o}^2}{4R_{eq}} = \frac{(2\sqrt{2})^2}{4 \times \frac{1}{2}} \text{ W} = 4 \text{ W}$$

【例17】

图示电路，已知 $f_1 = 100\text{kHz}$ 时，电流不能通过负载 R_L ，而在频率为 $f_2 = 50\text{kHz}$ 时流过 R_L 的电流为最大。求 C_1 和 C_2 。

【解】 L 和 C_1 发生**并联谐振**时，电流不能通过负载。



$$\text{则有 } \omega_1 C_1 = \frac{1}{\omega_1 L}$$

$$\Rightarrow C_1 = \frac{1}{\omega_1^2 L} = \frac{1}{(2\pi f_1)^2 \times 10 \times 10^{-6}} \approx 0.25\mu\text{F}$$

$L // C_1$ 和 C_2 发生**串联谐振**时，电流最大

$$\frac{1}{j\omega_2 C_2} + \frac{j\omega_2 L \cdot \frac{1}{j\omega_2 C_1}}{j\omega_2 L + \frac{1}{j\omega_2 C_1}} = 0 \Rightarrow C_2 = \frac{1}{\omega_2^2 L} - C_1 \approx 0.76\mu\text{F}$$

【例18】

已知： $U_l = 380\text{V}$

三相负载

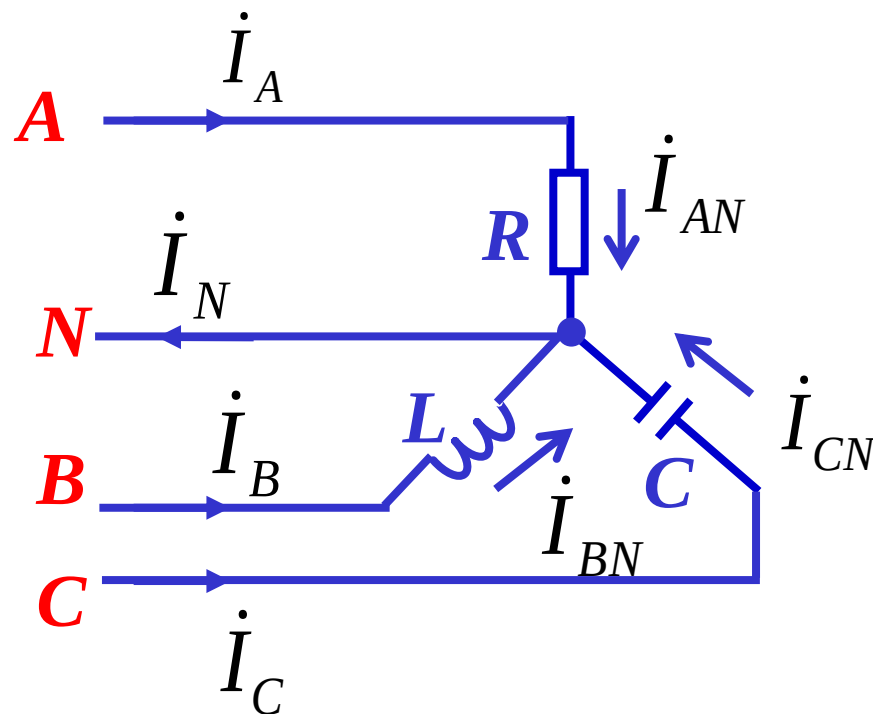
$$R = X_L = X_C = 10\Omega$$

求：相电流及中线电流

$$\dot{U}_{AN} = 220\angle 0^\circ \text{ V}$$

$$\dot{U}_{BN} = 220\angle -120^\circ \text{ V}$$

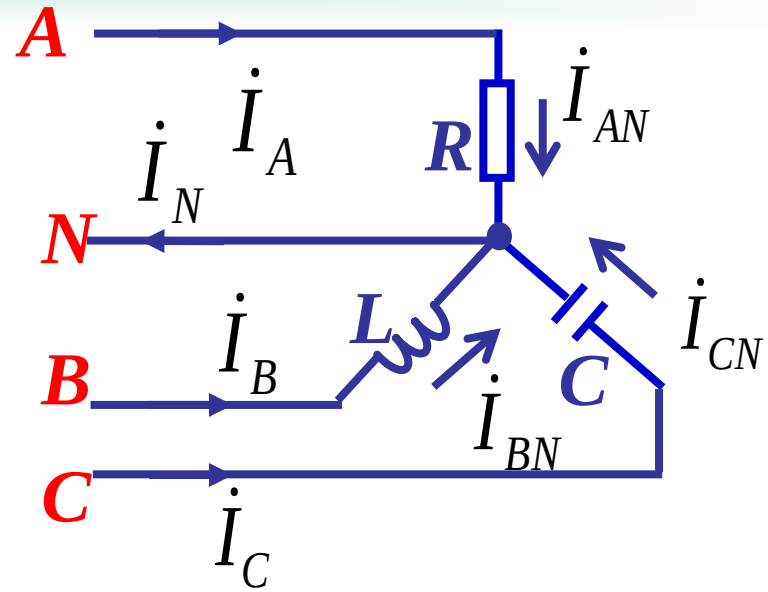
$$\dot{U}_{CN} = 220\angle 120^\circ \text{ V}$$



$$\dot{U}_{AN} = 220\angle 0^\circ \text{ V}$$

$$\dot{U}_{BN} = 220\angle -120^\circ \text{ V}$$

$$\dot{U}_{CN} = 220\angle 120^\circ \text{ V}$$

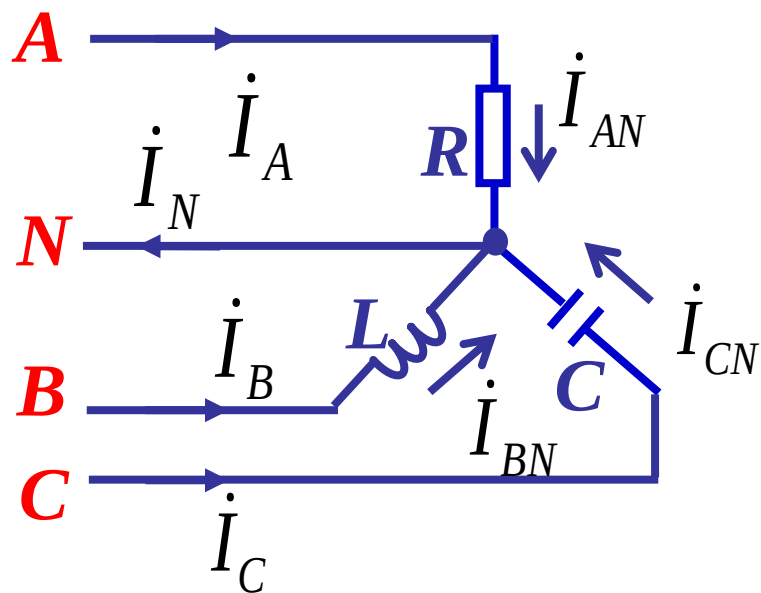
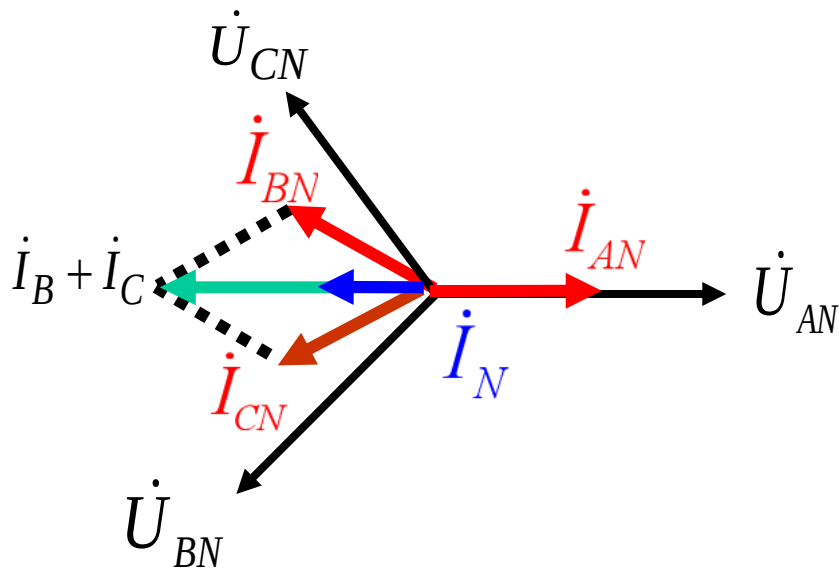


$$\left\{ \begin{array}{l} \dot{I}_{AN} = \frac{\dot{U}_{AN}}{R} = \frac{U_P}{R} = \frac{220}{10} = 22\angle 0^\circ \text{ A} \\ \dot{I}_{BN} = \frac{\dot{U}_{BN}}{j\omega L} = \frac{U_P\angle -120^\circ}{j\omega L} = \frac{220}{10}\angle -210^\circ = (22\angle -210^\circ) \text{ A} \\ \dot{I}_{CN} = \frac{\dot{U}_{CN}}{-j\frac{1}{\omega C}} = \frac{U_P\angle 120^\circ}{-j\frac{1}{\omega C}} = \frac{220}{10}\angle 210^\circ = (22\angle -150^\circ) \text{ A} \end{array} \right.$$

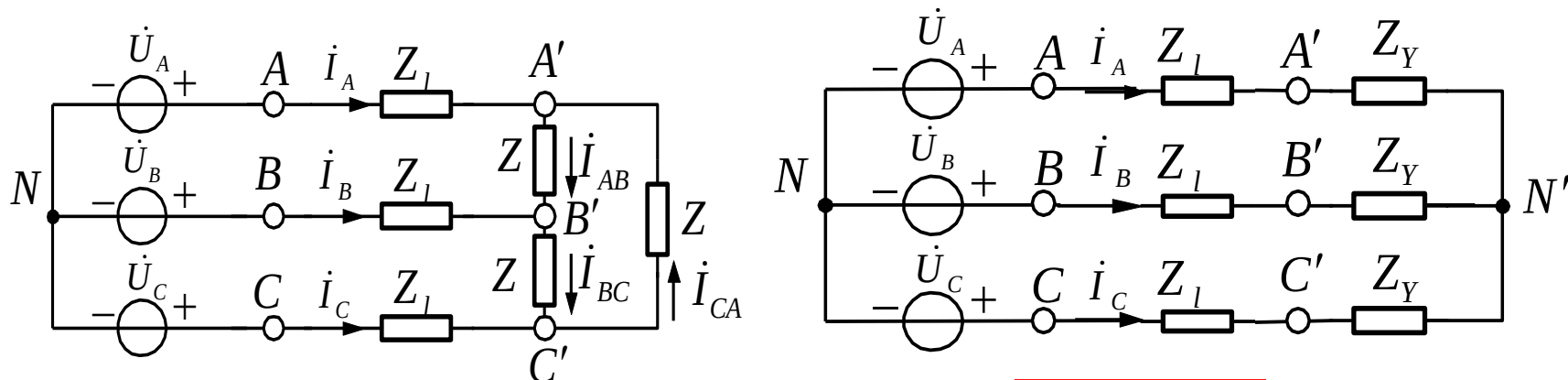
中线电流

$$\begin{aligned}\dot{I}_N &= \dot{I}_A + \dot{I}_B + \dot{I}_C \\ &= 22 + 22\angle -210^\circ + 22\angle -150^\circ \\ &= 22 - 19 - j11 - 19 + j11 \\ &= 22 - 38 = -16\text{ A}\end{aligned}$$

用相量图求中线电流



【例19】 已知电路如图所示, $Z = (19.2 + j14.4)\Omega$, $U_{AB} = 380V$ 。
 $Z_l = (3 + j4)\Omega$, 试求负载 Z 的相电压和相电流。

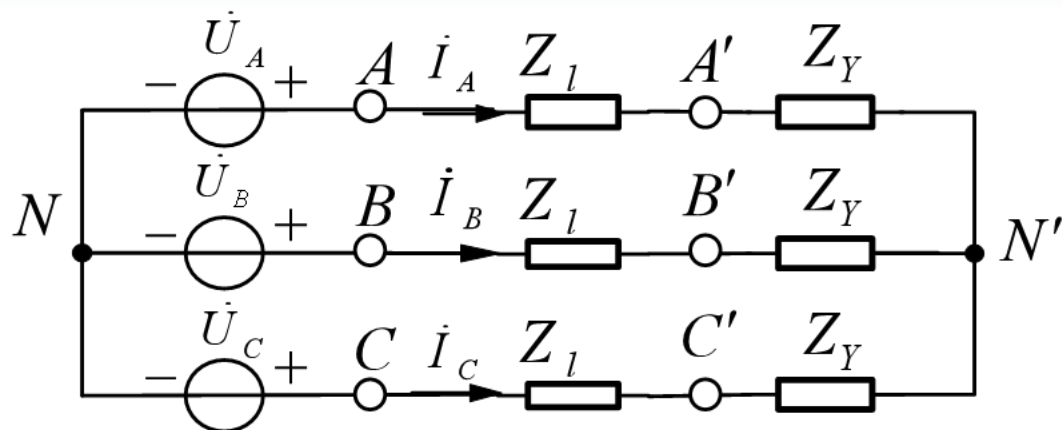


解: 将电路变换为对称的Y——Y电路 $Z_{\Delta} = 3Z_Y$

$$Z_Y = \frac{Z_{\Delta}}{3} = \frac{19.2 + j14.4}{3} = (6.4 + j4.8)\Omega \quad \dot{U}_A = 220\angle 0^\circ \text{ V}$$

$$\dot{I}_A = \frac{\dot{U}_A}{Z_l + Z_Y} = \frac{220\angle 0^\circ}{3 + j4 + 6.4 + j4.8} = 17.1\angle -43.2^\circ \text{ A}$$

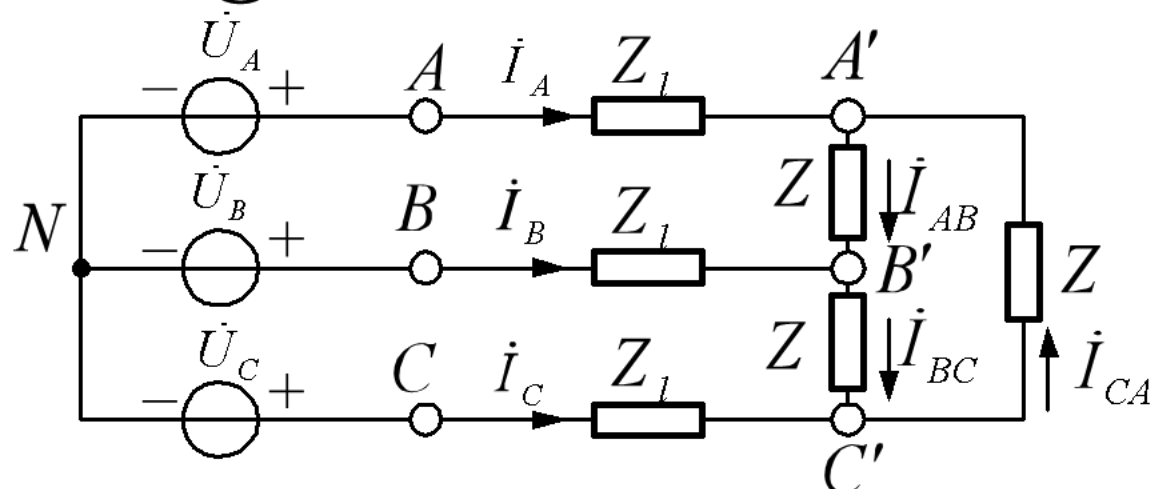
$$\dot{I}_B = 17.1\angle -163.2^\circ \text{ A} \quad \dot{I}_C = 17.1\angle 76.8^\circ \text{ A}$$



$$\dot{I}_A = 17.1 \angle -43.2^\circ \text{ A}$$

$$\dot{I}_B = 17.1 \angle -163.2^\circ \text{ A}$$

$$\dot{I}_C = 17.1 \angle 76.8^\circ \text{ A}$$



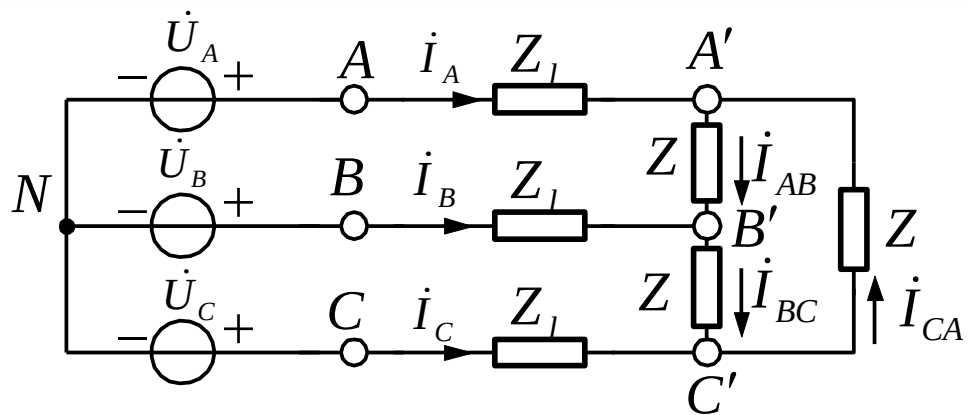
$$\dot{I}_l = \sqrt{3} \dot{I}_p \angle -30^\circ$$

$$Z = (192 + j144) \Omega$$

$$\dot{I}_{AB} = \frac{\dot{I}_A}{\sqrt{3} \angle -30^\circ} = \frac{17.1 \angle -43.2^\circ}{\sqrt{3} \angle -30^\circ} = 9.9 \angle -13.2^\circ \text{ A}$$

$$\dot{I}_{BC} = 9.9 \angle -132.2^\circ \text{ A}$$

$$\dot{I}_{CA} = 9.9 \angle 106.8^\circ \text{ A}$$



$$Z = (19.2 + j14.4)\Omega$$

相电流:

$$\dot{I}_{AB} = \frac{\dot{I}_A}{\sqrt{3} \angle -30^\circ} = \frac{17.1 \angle -43.2^\circ}{\sqrt{3} \angle -30^\circ} = 9.9 \angle -13.2^\circ \text{ A}$$

$$\dot{I}_{BC} = 9.9 \angle -132.2^\circ \text{ A}$$

$$\dot{I}_{CA} = 9.9 \angle 106.8^\circ \text{ A}$$

相电压:

$$\dot{U}_{A'B'} = Z \dot{I}_{AB} = (19.2 + j14.4) \times 9.9 \angle -13.2^\circ = 237.6 \angle 23.7^\circ \text{ V}$$

$$\dot{U}_{B'C'} = 237.6 \angle -96.3^\circ \text{ V}$$

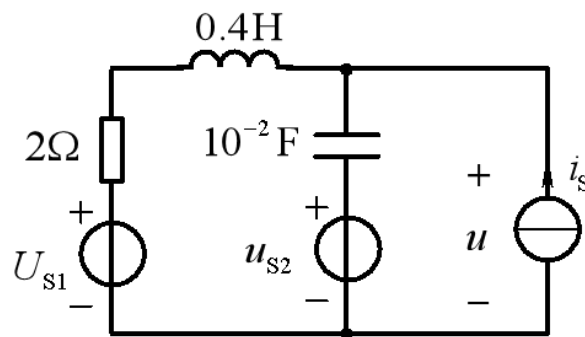
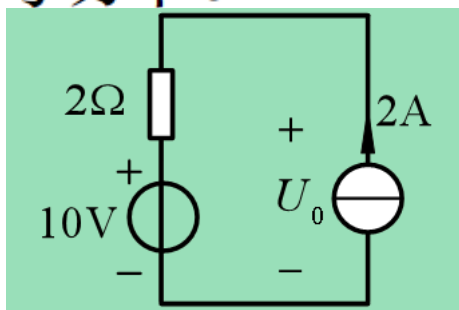
$$\dot{U}_{C'A'} = 237.6 \angle 143.7^\circ \text{ V}$$

【例20】

图示电路 $U_{s1} = 10V$, $u_{s2} = 20\sqrt{2} \cos \omega_1 t V$, $i_s = (2 + 2\sqrt{2} \cos \omega_1 t) A$
 $\omega_1 = 10 \text{ rad/s}$ 。 (1) 求电流源的端电压 u 及其有效值;
 (2) 求电流源发出的平均功率。

【解】 直流分量作用:

$$U_0 = 10V + 2\Omega \times 2A = 14V$$

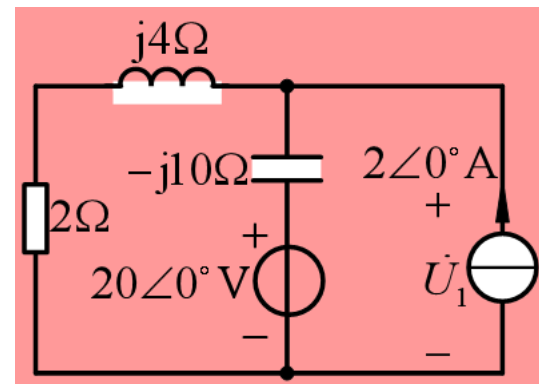


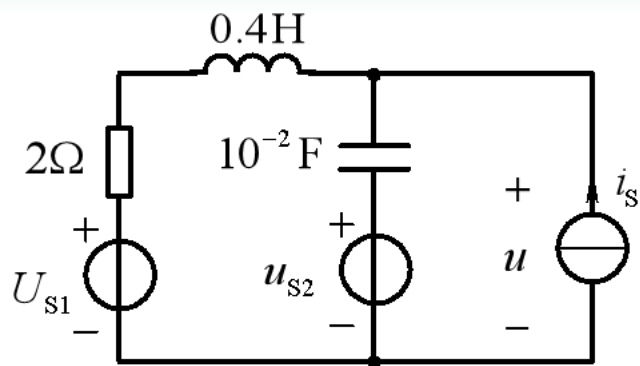
交流分量作用:

$$\dot{U}_1 = \frac{\frac{20}{-j10} + 2}{\frac{1}{2 + j4} + \frac{1}{-j10}}$$

解得

$$\dot{U}_1 = 20 \angle 90^\circ V$$





$$U_0 = 14\text{V}$$

$$\dot{U}_1 = 20\angle 90^\circ \text{ V}$$

电流源的端电压及其有效值分别为

$$u = U_0 + u_1 = [14 + 20\sqrt{2} \cos(\omega_1 t + 90^\circ)]\text{V}$$

$$U = \sqrt{U_0^2 + U_1^2} = \sqrt{(14)^2 + (20)^2} \text{ V} = 24.4\text{V}$$

电流源发出的平均功率

$$\begin{aligned} P &= 2U_0 + 2U_1 \cos(90^\circ - 0^\circ) \\ &= (14 \times 2 + 20 \times 2 \cos 90^\circ) \text{ W} = 28\text{W} \end{aligned}$$

练习结束